

HLD - Scalable Online Coding Assessment Platform

Functional Requirements

- Register/login
- View assessments
- Join assessment
- Submit code
- View Submission status
- View result

Students can join assessments, view assessment and submit code, view submission status and results. Admins can create assessments and view results.

Non-Functional Requirements

Scalable: Supports thousands of students.

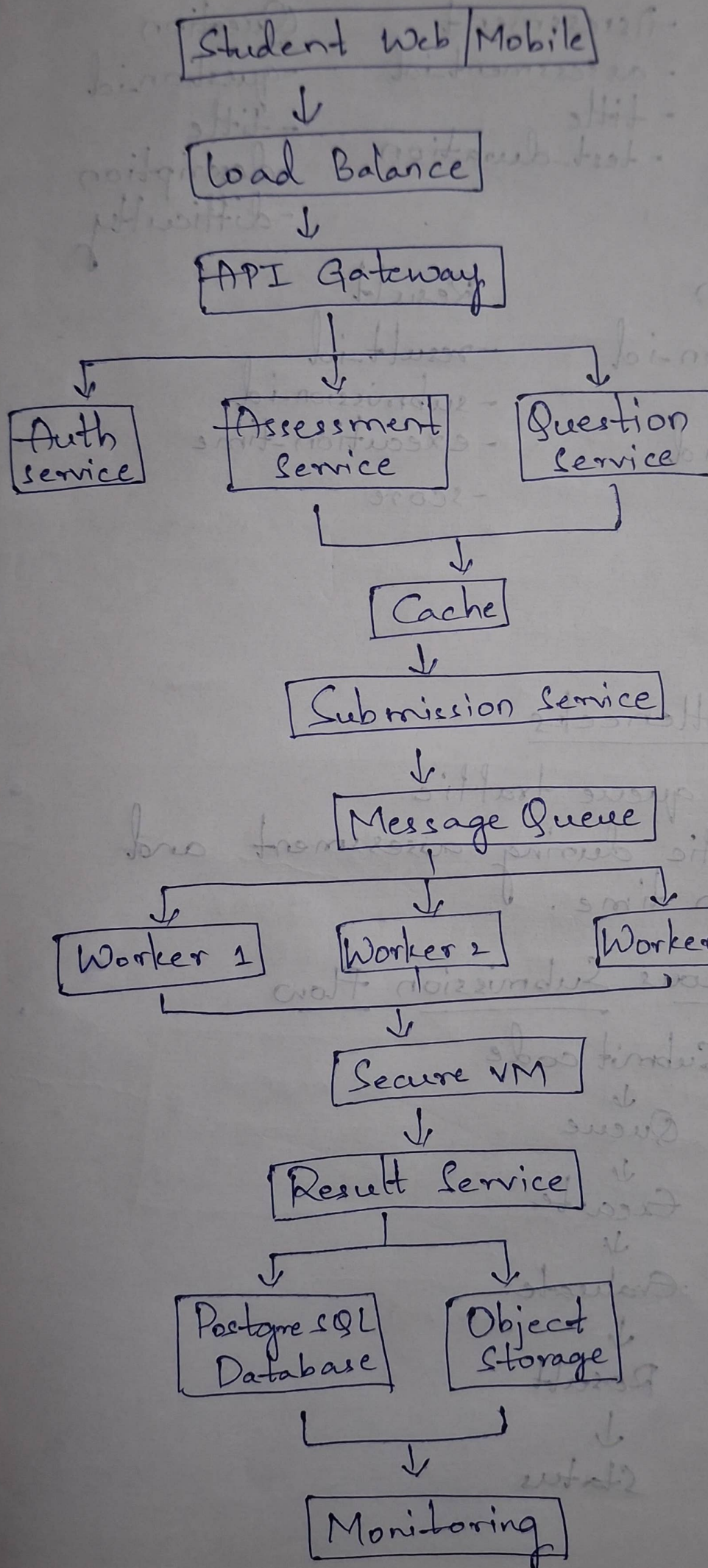
Available: System have to work without failures.

Secure: Execute code safely.

Reliable: Retry failed submission.

(Fast): Use caching & asynchronous processing.

Architecture



Entities

User

- user-id
- name
- email
- password

Tables

- Assessment
- assessment-id
- title
- test-duration

Question

- question-id
- title
- description
- difficulty

Submission

- submission-id
- user-id
- question-id
- language
- code
- status

Result

- result-id
- submission-id
- execution-time
- score

Main Bottlenecks

- Message queue traffic

API traffic during assessment and submission time.

Asynchronous Submission Flow

Submit code

↓
Queue

↓
Execute

↓
Evaluate

↓
Result

↓
Status

API list

POST/api/auth/login

GET/api/assessments

POST/api/assessments

GET/api/questions

POST/api/questions

GET/api/submissions/result

GET/api/submissions

POST/api/submissions

